



Zoé Julliard-Bonnouvriée

M2 Student in Data Science for Social Sciences — Toulouse School of Economics

Data Scientist — Seeking a 6-month end-of-studies internship (starting April 2027)

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ABOUT ME

Strongly grounded in a dual background in mathematics and data science, I design high-performance algorithmic solutions applied to complex systems (Agentic AI, Computer Vision, Time Series). I am seeking a 6-month end-of-studies internship starting in **April 2027** to leverage this dual expertise on projects in the industrial or banking sectors.

TECHNICAL SKILLS

Languages & Programming: Python (Advanced), R (Advanced), SQL (Intermediate), Git, LaTeX
Generative AI & MLOps: Agentic AI (Multi-agent), MCP, LLM, RAG, FastAPI, Dataiku DSS
Data Science & Machine Learning: Scikit-Learn, Pandas, NumPy, XGBoost, Random Forest, Segmentation
Computer Vision & Geospatial: PyTorch, U-Net, Transfer Learning (ResNet)
Languages: French (Native, C2), English (Fluent, C1 - TOEFL 99/120), Spanish (Basic, A2)

PROFESSIONAL EXPERIENCE

- Data Scientist & AI Engineer Intern – AVISIA (Data & AI Consulting Firm, Paris)** 2026
- Developed an Agentic AI Assistant to automate SAS code migration to Dataiku.
 - Implemented 5 specialized autonomous agents (using MCP, LLM, FastAPI) featuring a closed-loop error self-correction mechanism (dry-run, log inspection) and a dedicated UX interface.
 - Evaluated data parity between Dataiku and SAS Viya (schema and aggregate comparison testing).
 - Impact: Reduced migration time from several days to 5 minutes per script ($\sim \$0.15$ per run).
- Head of Treasury – TSE Junior-Entreprise** 2025–2027
- Managed the annual budget, oversaw accounting operations, and submitted financial tax declarations.
- Logistics Agent – Liebherr Aerospace (Toulouse)** 2024 & 2025
- Prepared industrial orders: organization, rigor, teamwork, and task prioritization.

DATA SCIENCE ACADEMIC PROJECTS

- Computer Vision – Semantic Segmentation of Satellite Imagery** Python, PyTorch, U-Net
- Developed a pixel-by-pixel segmentation pipeline for 7 land cover classes.
 - Preprocessed with 256×256 patches and trained a U-Net model with a pre-trained ResNet34 backbone (Transfer Learning).
 - Evaluated performance using *IoU* and *Dice Score*.
- Multi-Model Predictive Modeling on Large Datasets** R, Tidyverse
- Analyzed and cleaned large datasets including outlier detection and handling.
 - Implemented and trained regression models (Linear Regression, Random Forest, XGBoost).
 - Evaluated predictive accuracy through error metric comparisons (*RMSE*, R^2).
- Financial Time Series Analysis & Efficient Market Hypothesis** R, Forecast, Tseries
- Conducted stationarity tests (ADF, KPSS), ARIMA(0,0,1) modeling, and ADL models.
 - Performed residual white noise diagnostics (Ljung-Box) and detected heteroskedasticity/ARCH effects.

EDUCATION

- Master's Degree in Data Science for Social Sciences – International Track – Toulouse School of Economics** 2025–2027
- Coursework: Machine Learning, Deep Learning, Time Series, Advanced Econometrics, Inferential Statistics.
- Academic Exchange Year – University of Oslo (Norway)** 2024–2025
- Coursework: R, Data Science, Time Series, Econometrics, Corporate Finance.
- Dual Bachelor's Degree in Economics and Mathematics – TSE, Toulouse Capitole University** 2021–2024

INTERESTS

Handball (8 years) | Swimming (15 years) | Weightlifting (5 years) | Traveling Worldwide | Volunteering (US Aucamville Handball)